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State transformation-based dynamic visual servoing for an unmanned aerial vehicle

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ABSTRACT
In this paper, we propose a visual servoing control for a quadrotor unmanned aerial vehicle (UAV) which is based on a state transformation technique. The UAV is equipped with a single downwards facing camera, and the motion control objective is the regulation of relative displacement and yaw to a stationary visual target located on the ground. The state transformation is defined by a system of partial differential equations (PDEs) which eliminate roll and pitch rate dependence in the transformed image feature kinematics. A method for computing the general solutions of these PDEs is given, and we show a particular solution reduces to an established virtual camera approach. We treat point and line cases and introduce image moment features defined in the virtual camera image plane. Robustness of the control design is improved by accounting for attitude measurement bias, and uncertainty in thrust gain, mass, and image feature depth. The asymptotic stability of the closed-loop is proven. The method is based on a simple proportional-integral-derivative (PID) structure which can be readily implemented on-board. Experimental results show improved performance relative to previous work.

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State transformation; partial differential equations; image-based visual servoing; unmanned aerial vehicles (UAVs)

1. Introduction

Visual servoing involves using computer vision feedback to help guide a robot's motion (Chaumette & Hutchinson, 2006). Recently, there has been a particular interest in applying visual servoing to motion control of unmanned aerial vehicles (UAVs). Visual servoing can allow a UAV to explore locations where global positioning system (GPS) is unavailable (Fraundorfer et al., 2012) or solve a specific relative positioning task such as automated landing (Ma, Soatto, Kosecka, & Sastry, 2003). In this paper, we focus on visual servoing for a quadrotor UAV with a single onboard downward facing camera, and the control objective is to regulate the UAV's relative displacement and yaw to a visual target defined by point or line features and located on the ground. Conventionally, a visual seroving law relies on kinematics of an error signal which depends on an image measurement. The resulting control structure is divided into two feedback loops (Chaumette & Hutchinson, 2006). An outer loop uses low-rate video camera feedback to compute a velocity reference based on the kinematics, and a high-bandwidth inner loop tracks this reference. This two-loop structure is natural due to the difference in sampling rates of the loops. The structure simplifies the control design by assuming perfect inner-loop tracking. However, as pointed out in Chaumette and Hutchinson (2007) and Jabbari, Oriolo, and Bolandi (2014), when high-performance motion control is required or the system is underactuated, the dynamics of the robot should be considered. *Dynamic* visual servoing directly accounts for the robot's dynamics and is a challenging problem due to the nonlinearity of the model. Visual servoing is normally categorised into two basic approaches: *position-based visual servoing* and *image-based visual servoing* (IBVS). PBVS methods reconstruct the vehicle pose from image measurements and control the motion directly in SE(3), e.g. (Azrad, Kendoul, & Nonami, 2010; Mejias, Saripalli, Campoy, & Sukhatme, 2006; Shakernia, Ma, Koo, & Sastry, 1999). A benefit of PBVS is that non-vision based motion controllers can be used. In this paper, we take an IBVS approach where the feature error is directly computed from the image measurements. This avoids three-dimensional (3D) position reconstruction which requires accurate camera calibration and a prior 3D model of a target. In general, IBVS is relatively insensitive to camera calibration error even though the IBVS law can depend on the camera's intrinsic parameters (Espiau, 1994). Another feature of IBVS is that the target object tends not to leave the field of view (FoV) since the design minimises image feature error. However, the image feature kinematics in IBVS

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is usually nonlinear and this complicates the control design.

A number of IBVS approaches have been proposed for UAVs. The seminal work in Hamel and Mahony (2002) shows that a perspective projection camera model destroys the triangular structure of the system dynamics. Hence, a spherical projection camera model is introduced to give the image feature kinematics a passivity-like property, i.e., the time derivative of the norm of the error function is independent of angular velocity. By introducing first-order moment point features, a backstepping controller is proposed. As the features in Hamel and Mahony (2002) are insensitive to vertical motion due to an ill-conditioned interaction matrix, work in Bourquardez, Mahony, Hamel, and Chaumette (2006) modifies them by rescaling and partitioning. In Bourquardez et al. (2009), the modified image moment features are applied to visual servoing of a UAV. A similar approach to Hamel and Mahony (2002) is in Mahony and Hamel (2005) where first-order line image moment features are proposed for applications with linear structure such as following power lines or pipelines. We remark that in the spherical projection approach, it is hard to propose a feature to control yaw. It is well known that a homography matrix embeds information about the relative pose between two camera frames (Ma et al., 2003). Hence, in Metni, Hamel, and Derkx (2005), de Plinval, Morin, Mouyon, and Hamel (2011), and de Plinval, Morin, Mouyon, and Hamel (2013) a visual servoing scheme is proposed using image features extracted from the homography matrix which maps image points between the desired and current image planes. Compared to approaches using spherical moment features, this approach only works for planar targets. Work in Metni et al. (2005), de Plinval et al. (2011), and de Plinval et al. (2013) either controls partial motion of the UAV or is limited to a small range of image feature error. With the assumption that the image and target planes remain parallel during motion of the vehicle, point moment features and a virtual spring concept lead to a design which accounts for the non-passive property of the kinematics (Ozawa & Chaumette, 2011, 2013). The assumption in this approach is potentially restrictive for more aggressive UAV maneuvers since lateral motion leads to non-parallel image and target planes. The attitude of a UAV is commonly estimated using an attitude and heading reference system (AHRS) which is integrated into most UAV autopilots. Measurement of a vehicle's attitude makes it possible to define a virtual camera with zero roll and pitch which can be used for visual servoing, e.g. Jabbari et al. (2014), Lee, Lim, H., Kim, and Seong (2012), and Xie, Lynch, and Jagersand (2014). In the virtual camera approach, features used for visual servoing are computed in the virtual camera frame. The image kinematics is simpler since it is independent of roll and pitch rates. In Xie et al. (2014), an adaptive IBVS controller for linear structure is proposed based on a virtual camera. In Abdessameud and Janabi-Sharifi (2015), a virtual camera with no rotational motion is proposed and an output feedback controller from Abdessameud and Tayebi (2010) is modified to achieve visual servoing for a moving target. Le Bras, Hamel, Mahony, and Treil (2011) utilise a virtual plane which is parallel to the target plane and tangent to the unit sphere in the spherical projection. The area of the convex hull of the image points in the virtual camera and the first-order image moment are used to form the image feature for visual servoing. An adaptive output feedback backstepping controller is developed which does not require velocity measurement.

In this paper, we adopt a different two-loop structure for dynamic visual servoing. The outer loop design is based on the UAV translational dynamics not its kinematics. The inner loop tracks the reference attitude signal from the outer loop. The control design is complicated due to the appearance of the derivative of attitude in the outer loop. To eliminate this dependence in the outer loop, we introduce new state coordinates. The new state is defined by a system of first-order linear homogeneous partial differential equations (PDEs) whose general solution can be obtained. We show the virtual camera approach proposed in Jabbari et al. (2014), Lee et al. (2012), and Xie et al. (2014) is a particular solution to the PDEs. Hence, the proposed scheme generalises the virtual camera approach. A constant rotation is used to generalise the virtual camera in Jabbari et al. (2014), Lee et al. (2012), and Xie et al. (2014). We introduce commonly used moment features to simplify the image feature kinematics so that they have linear dependence on velocity. The proposed control is robust to system uncertainty in thrust gain, mass, and image feature depth. We also account for attitude measurement bias which leads to non-zero steady-state error as shown in Fink, Xie, Lynch, and Jagersand (2015). For simplicity of on-board implementation, our approach is based on a proportional-integral-derivative (PID) structure. We achieve the same level of robustness as more complex adaptive methods such as Jabbari et al. (2014), Xie et al. (2014). Another important contribution of our work lies in its experimental validation. This should be compared to Jabbari et al. (2014) and Xie et al. (2014) where only simulation results are provided. To the best of the authors' knowledge, this is the first experimental validation of visual servoing with a virtual camera for line structures.

The paper is organised as follows. Section 2 describes the quadrotor dynamics, basic point and line image measurements and their kinematics. Section 3 describes the

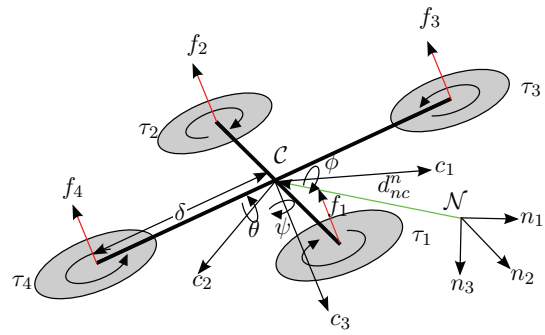


Figure 1. Quadrotor modelling.

state transformation scheme and presents image moment features in the virtual camera. We present the system uncertainty and robust dynamic IBVS law in Section 4. Experimental validation is given in Section 5.

2. Modelling

In this section, we introduce the dynamic model of a quadrotor used for dynamic IBVS. Basic image features for points and lines and their kinematics are described.

2.1. Quadrotor dynamics

As shown in Figure 1, two reference frames are used to describe the dynamics of a quadrotor. A navigation frame $\mathcal{N}$ is assumed to be inertial and is fixed to a point on the Earth with a basis $\{n_1, n_2, n_3\}$ whose elements are oriented north, east, and down, respectively. A downward looking pinhole camera is fixed at the centre of mass of the vehicle, and a camera frame $\mathcal{C}$ is defined with its origin fixed at the camera's optical centre and has a basis $\{c_1, c_2, c_3\}$ oriented forward, right, and down, respectively. The vector c_3 is aligned with the optical axis. The vector $d_{nc}^c = [d_1, d_2, d_3]^T$ denotes the displacement of the vehicle in $\mathcal{N}$. The dynamics of the UAV is

$$\dot{v}^c = -[\omega^c]_\times v^c + gR^T E_3 + F^c/m \tag{1a}$$

$$\dot{R} = R[\omega^c]_\times \tag{1b}$$

$$\dot{\omega}^c = -J^{-1}[\omega^c]_\times J\omega^c + J^{-1}\tau^c \tag{1c}$$

where v^c, ω^c are the linear and angular velocity of the camera in $\mathcal{C}$, respectively, $[a]_\times b = a \times b$, g is the gravity constant, m is the mass of the vehicle, J is the inertia of the vehicle, $E_3 = [0, 0, 1]^T$, F^c is the external force expressed in $\mathcal{C}$, τ^c is the external torque expressed in $\mathcal{C}$, and R is the rotation matrix which represents the relative orientation of $\mathcal{N}$ and $\mathcal{C}$. The rotation matrix R is parametrised by the ZYX Euler angles $\eta = [\phi, \theta, \psi]^T$ as $R = R_{n_3}(\psi)R_{n_2}(\theta)R_{n_1}(\phi)$ where R_{n_k} denotes an elementary rotation about the n_k axis. We remark that the Euler angle parametrisation of the rotation matrix has singularities. However, singularities are normally only encountered in acrobatic flight which is not typical in visual servoing where visual targets must remain in the camera's FoV. Euler angles are a common choice in field-tested autopilots, e.g. (Godbolt, 2013; Lim, Park, Lee, & Kim, 2012; PX4, 2014). The angular velocity ω^c is related to the Euler angle rate by

$$\omega^c = W(\eta)\,\dot{\eta} \tag{2}$$

where

$$W(\eta) = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix}$$

and $c_\xi = \cos\xi$, $s_\xi = \sin\xi$. Ignoring small body force, gyroscopic force, motor system inertia, we approximate the external force as $F^c = E_3 T_M$, where $T_M = \sum_{i=1}^4 f_i$, and f_i is the thrust created by the ith propeller. Each motor is driven by an electronic speed controller (ESC) that takes a Pulse Width Modulation (PWM) signal W_i as its input. We approximate the relation between f_i and W_i as $f_i = K_T \tilde{W}_i^2$, where $\tilde{W}_i = W_i - W_{\min}$, $W_{\min}$ is the minimum pulse width to rotate the propeller, and K_T is the thrust coefficient. We approximate τ^c as

$$\tau^c = \begin{bmatrix} \delta\left(f_2 + f_3 - f_1 - f_4\right)/\sqrt{2} \\ \delta\left(f_1 + f_3 - f_2 - f_4\right)/\sqrt{2} \\ -K_\tau\left(\tilde{W}_1^2 + \tilde{W}_2^2 - \tilde{W}_3^2 - \tilde{W}_4^2\right) \end{bmatrix}$$

where δ is the distance from the motor to the centre of mass of the vehicle, and K_τ is an aerodynamic constant.

2.2. Basic image measurements and their image kinematics

In this section, we describe two basic image measurements for points and lines. These measurements will be combined to form an m-vector image feature $s \in \mathbb{R}^m$. In this paper, we assume the target is static and so the trajectory of s depends only on camera motion. The image feature kinematics can be written as

$$\dot{s} = L_v v^c + L_\omega \omega^c \tag{3}$$

where L_v and L_ω are the image Jacobians or interaction matrices. The measurement for a 3-D point target P with coordinates $[X_1, X_2, X_3]^T$ in $\mathcal{C}$ is its image coordinates $p = [p_1, p_2]^T = [\lambda X_1/X_3, \lambda X_2/X_3]^T$, where λ is the focal length of the camera. As in Chaumette and Hutchinson (2006),

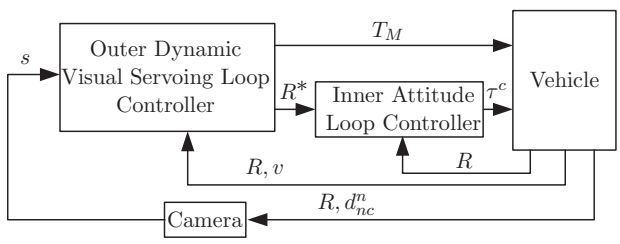


Figure 2. Inner–outer control loop structure.

the kinematics of p is

$$\dot{p} = L_{vp}v^c + L_{\omega p}\omega^c \tag{4}$$

where $L_{vp} = \begin{bmatrix} -\lambda/X_3 & 0 & p_1/X_3 \\ 0 & -\lambda/X_3 & p_2/X_3 \end{bmatrix}$, $L_{\omega p} = \begin{bmatrix} p_1p_2/\lambda & -(p_1^2+\lambda^2)/\lambda & p_2 \\ (p_2^2+\lambda^2)/\lambda & -p_1p_2/\lambda & -p_1 \end{bmatrix}$. A 3D straight line denoted as ℓ expressed in $\mathcal{C}$ is the intersection of two planes

$$h_i^T P + D_i = A_iX_1 + B_iX_2 + C_iX_3 + D_i = 0, \quad i = 1, 2$$

where $h_i = [A_i\, B_i\, C_i]^T$ is the normal vector of ith plane expressed in $\mathcal{C}$. The projection of ℓ can be parametrised by $l = [l_1, l_2]^T = [\alpha, \rho]^T$ as $p_1s_\alpha + p_2c_\alpha - \rho = 0$, where p_1 and p_2 are the image coordinates of a point in the line. As in Xie et al. (2014), the kinematics of l is

$$\dot{l} = L_{vl}v^c + L_{\omega l}\omega^c \tag{5}$$

where $L_{vl} = \begin{bmatrix} \mu_\alpha\lambda s_\alpha & \mu_\alpha\lambda c_\alpha & -\mu_\alpha\rho \\ \mu_\rho\lambda s_\alpha & \mu_\rho\lambda c_\alpha & -\mu_\rho\rho \end{bmatrix}$, $L_{\omega l} = \begin{bmatrix} \rho s_\alpha/\lambda & \rho c_\alpha/\lambda & 1 \\ (\lambda+\rho^2/\lambda)c_\alpha & -(\lambda+\rho^2/\lambda)s_\alpha & 0 \end{bmatrix}$, and $\mu_\alpha = -(A_ic_\alpha - B_is_\alpha)/(D_i\lambda)$, $\mu_\rho = (A_i\rho s_\alpha + B_i\rho c_\alpha + C_i\lambda)/(D_i\lambda)$.

3. The state transformation approach

In this section, we justify the inner–outer loop controller structure used and explain how a state transformation approach eliminates input derivatives in the outer-loop. The approach is illustrated on a simple planar vertical take-off and landing (PVTOL) vehicle. Next, we consider point and line basic measurements, introduce image moment features in the new states, and provide the resulting kinematics.

3.1. General philosophy

As discussed in Section 1, the classical IBVS approach cannot be applied to an underactuated UAV where the velocities v^c and ω^c are not independently controlled. In this paper, we consider a different two-loop structure as shown in Figure 2. The inner loop tracks an attitude reference denoted R^*, and the outer-loop provides this reference to the inner loop in order to regulate the image feature error $e_s = s - s^*$, where s denotes the image feature to be chosen below and s^* denotes the desired value of s. A main reason for adopting this control structure is that it provides a reliable and simple solution for motion control. Each loop becomes fully actuated and this simplifies the control design and gain tuning to optimise performance. A similar two-loop structure for conventional position control is found in many autopilots, e.g. Godbolt (2013), PX4 (2014). Stability of the closed-loop system requires the inner loop to have a relatively high bandwidth. The visual servoing problem naturally fits the inner–outer loop structure since AHRS measurements are relatively fast compared to image feedback. Furthermore, the inner loop uses a relatively high-gain control to ensures closed-loop stability. We discuss the conditions for stability in Section 4.4.3. One alternative is to use ω^c as the control for the outer loop, e.g. Guenard, Hamel, and Mahony (2008). However, a benefit of the proposed structure is its direct control of attitude which ensure attitude remains in a safe region. It also makes it easier to keep the target in the camera's FoV.

If we take the inner–outer loop structure as shown in Figure 2 for visual servoing, the outer-loop dynamics (1a) and (3) contains the attitude input and its derivative. The appearance of the input derivative makes the control design challenging. One way to solve this problem is to lower the input derivative by a state transformation. This approach was first proposed in Fink et al. (2015) for visual servoing of lateral motion. This paper extends the work in Fink et al. (2015) to the general motion of the UAV. From (1a), (2), and (3), the outer loop is

$$\dot{s} = L_v v^c + L_{\eta_1}\dot{\eta}_1 + L_{\eta_2}\dot{\psi} \tag{6a}$$

$$\dot{v}^c = H_u + H_{\eta_1}\dot{\eta}_1 + H_{\eta_2}\dot{\psi} \tag{6b}$$

where $[L_{\eta_1}, L_{\eta_2}] = L_\omega W$, $H_u = gR^TE_3 + F^c/m$, $[H_{\eta_1}, H_{\eta_2}] = [v^c]_\times W$, and $\eta_1 = [\phi, \theta]^T$. The control variables for (6) are F^c, η_1, and ψ. The yaw angle ψ can be controlled independently. We define a state transformation $s^a(s, \eta_1)$ and $v^a(v^c, \eta_1)$ with dependence on input η_1. The time derivative of the state transformation is

$$\dot{s}^a = \frac{\partial s^a}{\partial s}L_v v^c + \left(\frac{\partial s^a}{\partial s}L_{\eta_1} + \frac{\partial s^a}{\partial \eta_1}\right)\dot{\eta}_1 + \frac{\partial s^a}{\partial s}L_{\eta_2}\dot{\psi}$$
$$\dot{v}^a = \frac{\partial v^a}{\partial v^c}H_u + \left(\frac{\partial v^a}{\partial v^c}H_{\eta_1} + \frac{\partial v^a}{\partial \eta_1}\right)\dot{\eta}_1 + \frac{\partial v^a}{\partial v^c}H_{\eta_2}\dot{\psi}$$

If s^a and v^a satisfy the system of first-order linear homogeneous PDEs

$$\frac{\partial s^a}{\partial s} L_{\eta_1} + \frac{\partial s^a}{\partial \eta_1} = 0 \tag{8a}$$

$$\frac{\partial v^a}{\partial v^c} H_{\eta_1} + \frac{\partial v^a}{\partial \eta_1} = 0 \tag{8b}$$

the input derivative in (6) can be eliminated and the dynamics (6) expressed in the new state is

$$\dot{s}^a = \frac{\partial s^a}{\partial s} L_v v^c + \frac{\partial s^a}{\partial s} L_{\eta_2} \dot{\psi}$$
$$\dot{v}^a = \frac{\partial v^a}{\partial v^c} H_u + \frac{\partial v^a}{\partial v^c} H_{\eta_2} \dot{\psi}$$

3.2. Example: the PVTOL vehicle

In this section, we consider a PVTOL vehicle ignoring lateral coupling to illustrate the basic philosophy of the state transformation approach with a point measurement. The PVTOL dynamics is obtained by setting $\omega_1^c, \omega_3^c, v_2^c, \phi, \psi, p_2$ in (4) and (1a) to zero, we obtain the PVTOL vehicle model

$$\dot{p}_1 = \begin{bmatrix} -\lambda/X_3 & p_1/X_3 \end{bmatrix} \begin{bmatrix} v_1^c \\ v_3^c \end{bmatrix} - \frac{p_1^2 + \lambda^2}{\lambda} \omega_2 \tag{10a}$$

$$\begin{bmatrix} \dot{v}_1^c \\ \dot{v}_3^c \end{bmatrix} = -\omega_2 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} v_1^c \\ v_3^c \end{bmatrix} + g \begin{bmatrix} -s_\theta \\ c_\theta \end{bmatrix} + \begin{bmatrix} 0 \\ -T_M/m \end{bmatrix} \tag{10b}$$

Relative to the 3D UAV model, (10) has fewer states, but keeps the essential structure of (4) and (1a).

The PDEs (8) become

$$\frac{\partial p_1^a}{\partial p_1} \frac{p_1^2 + \lambda^2}{\lambda} - \frac{\partial p_1^a}{\partial \theta} = 0, \quad \frac{\partial v_i^a}{\partial v_1^c} v_3^c - \frac{\partial v_i^a}{\partial v_3^c} v_1^c - \frac{\partial v_i^a}{\partial \theta} = 0 \tag{11}$$

where $i = 1, 3$. The general solutions of (11) are

$$p_1^a = \kappa_1(\arctan(p_1/\lambda) + \theta) \tag{12}$$

$$v_i^a = \kappa_{2i}\left(\left(v_1^c\right)^2 + \left(v_3^c\right)^2, \arctan\left(v_1^c/v_3^c\right) + \theta\right) \tag{13}$$

where κ_1 and κ_2 are smooth functions in their arguments. In Fink et al. (2015), κ_1 in (12) is chosen as $\kappa_1(\zeta) = \zeta$. By assuming $v_3^c = 0$ for the hovering case, the outer-loop in the new coordinates is

$$\dot{p}_1^a = v_1^c \cos^2\left(p_1^a - \theta\right)/X_3 \tag{14a}$$

$$\dot{v}_1^c = -g \sin\theta \tag{14b}$$

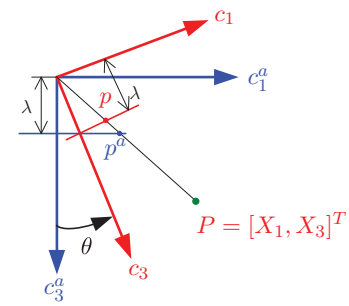


Figure 3. Virtual camera for the PVTOL example.

As in Fink et al. (2015), backstepping can used to stabilise system (14). If we choose $\kappa_1(\zeta) = \tan\zeta$, $\kappa_{21}(\zeta_1, \zeta_2) = \zeta_1 s_{\zeta_2}$, $\kappa_{22}(\zeta_1, \zeta_2) = \zeta_1 c_{\zeta_2}$, the state transformation is

$$p_1^a = \lambda(c_\theta p_1 + s_\theta \lambda)/(-s_\theta p_1 + c_\theta \lambda)$$
$$\begin{bmatrix} v_1^a \\ v_3^a \end{bmatrix} = \begin{bmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{bmatrix} \begin{bmatrix} v_1^c \\ v_3^c \end{bmatrix}$$

and the outer loop expressed in the new state is

$$\dot{p}_1^a = \frac{\lambda}{X_3\left(-s_\theta p_1 + c_\theta \lambda\right)} \begin{bmatrix} -\lambda & p_1^a \end{bmatrix} \begin{bmatrix} v_1^a \\ v_3^a \end{bmatrix} \tag{16a}$$

$$\begin{bmatrix} \dot{v}_1^a \\ \dot{v}_3^a \end{bmatrix} = g \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \frac{T_M}{m} \begin{bmatrix} s_\theta \\ c_\theta \end{bmatrix} \tag{16b}$$

As shown in Figure 3, we define a virtual camera frame $\mathcal{A}$ with basis $\{c_1^a, c_3^a\}$, which has the same translational motion as $\mathcal{C}$ but with zero pitch. Any point in $\mathcal{A}$ has coordinates

$$P^a = \begin{bmatrix} X_1^a \\ X_3^a \end{bmatrix} = \begin{bmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{bmatrix} P$$

The term $\lambda/(X_3(-s_\theta p_1 + c_\theta \lambda))$ in (16a) can be written as $1/(X_3 c_\theta - X_1 s_\theta)$ whose denominator is the depth X_3^a of the target in the virtual camera frame. Since the vehicle always hovers above the target $X_3^a > 0$. We also observe $p_1^a = \lambda \frac{X_3 s_\theta + X_1 c_\theta}{X_3 c_\theta - X_1 s_\theta} = \lambda \frac{X_1^a}{X_3^a}$, which is the image point in the virtual camera plane. Hence, the state transformation approach includes the virtual camera approach in Jabbari et al. (2014) as a special case.

3.3. State transformation for a 3D point and 3D line

The state transformations for the point feature case are denoted as $p_i^a(p_1, p_2, \phi, \theta)$, $v_i^a(v_1, v_2, \phi, \theta)$, $i = 1, 2$. From (4), (6), (8a), the new state p_i^a has to satisfy following PDEs

$$G_{jp}(p_i^a) = 0, \quad j, i = 1, 2 \tag{17}$$

where vector fields $G_{1p} = \frac{p_1 p_2}{\lambda}\frac{\partial}{\partial p_1} + \frac{p_2^2+\lambda^2}{\lambda}\frac{\partial}{\partial p_2} + \frac{\partial}{\partial \phi}$, $G_{2p} = \left(\frac{p_1^2+\lambda^2}{\lambda}c_\phi + p_2 s_\phi\right)\frac{\partial}{\partial p_1} + \left(\frac{p_1 p_2 c_\phi}{\lambda} - p_1 s_\phi\right)\frac{\partial}{\partial p_2} - \frac{\partial}{\partial \theta}$. As discussed in Appendix A, a solution to (17) exists and we follow the method presented there to compute a general solution

$$p_i^a = \kappa_{3i}\left(\frac{s_\phi c_\theta p_2 + \lambda c_\phi c_\theta - p_1 s_\theta}{\lambda s_\phi - p_2 c_\phi}, \frac{s_\phi s_\theta p_2 + c_\phi s_\theta \lambda + p_1 c_\theta}{\lambda s_\phi - p_2 c_\phi}\right) \tag{18}$$

where $i = 1, 2$, and κ_{3i} is an arbitrary smooth function in its arguments. The PDEs (8b) are

$$G_{jv}(v_i^a) = 0, \quad j = 1, 2, i = 1, 2, 3 \tag{19}$$

where $G_{1v} = v_3^c\frac{\partial}{\partial v_2^c} - v_2^c\frac{\partial}{\partial v_3^c} + \frac{\partial}{\partial \phi}$, $G_{2v} = -(v_3^c c_\phi + v_2^c s_\phi)\frac{\partial}{\partial v_1^c} + v_1^c s_\phi\frac{\partial}{\partial v_2^c} + v_1^c c_\phi\frac{\partial}{\partial v_3^c} + \frac{\partial}{\partial \theta}$. As with (17), we obtain the general solution

$$\begin{aligned} v_i^a = \kappa_{4i}(&v_2 c_\phi - v_3 s_\phi, v_1 c_\theta + v_2 s_\phi s_\theta + v_3 c_\phi s_\theta, \\ &-v_1 s_\theta + v_2 s_\phi c_\theta + v_3 c_\phi c_\theta) \end{aligned} \tag{20}$$

where $i = 1, 2, 3$, and κ_{4i} is an arbitrary smooth function. We take

$$\kappa_{31}(\zeta_1, \zeta_2) = \lambda\frac{E_1^T R_c q}{E_3^T R_c q}, \quad \text{and} \quad \kappa_{32}(\zeta_1, \zeta_2) = \lambda\frac{E_2^T R_c q}{E_3^T R_c q} \tag{21}$$

where $q = [\zeta_2, -1, \zeta_1]^T$, R_c is a constant rotation matrix, $E_1 = [1, 0, 0]^T$, $E_2 = [0, 1, 0]^T$, and

$$\kappa_{4i}(\zeta_1, \zeta_2, \zeta_3) = E_i^T R_c[\zeta_1, \zeta_2, \zeta_3]^T, \ i = 1, 2, 3 \tag{22}$$

The outer loop transformed in the new coordinates is

$$\begin{bmatrix} \dot{p}_1^a \\ \dot{p}_2^a \end{bmatrix} = L_{vp}^a v^a + L_{\eta_2}^a R_c^T E_3 \dot{\psi} \tag{23a}$$

$$\dot{v}^a = \left[\dot{\psi} R_c^T E_3\right]_\times v^a + F^a \tag{23b}$$

where $L_{vp}^a = \begin{bmatrix} -\lambda/X_3^a & 0 & p_1^a/X_3^a \\ 0 & -\lambda/X_3^a & p_2^a/X_3^a \end{bmatrix}$, $L_{\eta_2}^a = \begin{bmatrix} p_1^a p_2^a/\lambda & -((p_1^a)^2+\lambda^2)/\lambda & p_2^a \\ ((p_2^a)^2+\lambda^2)/\lambda & -p_1^a p_2^a/\lambda & -p_1^a \end{bmatrix}$ and $F^a = R_c R_{\theta\phi} g_1(F^c, \eta)$, $X_3^a = E_3^T R_c R_{\theta\phi} P$. The state transformations (21) and (22) generalise the work in Jabbari et al. (2014) to convert the point coordinates in the real camera image to the coordinates of a point projected to a virtual camera with zero roll and pitch. It can readily be shown that if R_c in (21) and (22) is replaced with $R_c R_{n_3}(\psi)$, the yaw rate $\dot{\psi}$ in (23) will disappear and this is the virtual camera used in Abdessameud and Janabi-Sharifi (2015).

The state transformation for a line feature is $l_i^a(\alpha, \rho, \phi, \psi)$. Using (5) and (6), (8a) becomes

$$G_{jl}(l_i^a) = 0, \quad j, i = 1, 2 \tag{24}$$

where $G_{1l} = \frac{\rho s_\alpha}{\lambda}\frac{\partial}{\partial \alpha} + \left(\lambda + \frac{\rho^2}{\lambda}\right)c_\alpha\frac{\partial}{\partial \rho} + \frac{\partial}{\partial \phi}$, $G_{2l} = \left(\frac{\rho c_\alpha}{\lambda}c_\phi - s_\phi\right)\frac{\partial}{\partial \alpha} - \left(\lambda + \frac{\rho^2}{\lambda}\right)s_\alpha c_\phi\frac{\partial}{\partial \rho} + \frac{\partial}{\partial \theta}$. Following the same procedure used to solve (17) and (19), we can obtain a general solution to (24). The particular solution similar to (21) is given by

$$l^a = \begin{bmatrix} \alpha^a \\ \rho^a \end{bmatrix} = \begin{bmatrix} \arctan(l_{e1}/l_{e2}) \\ -l_{e3}/\sqrt{l_{e1}^2 + l_{e2}^2} \end{bmatrix} \tag{25}$$

where

$$l_e = [l_{e1}, l_{e2}, l_{e3}]^T = BR_c R_{\theta\phi} B^{-1}[s_\alpha, c_\alpha, -\rho]^T \tag{26}$$

and $B = \text{diag}([0, 0, \lambda])$. According to (8), the state transformation for velocity v^a should satisfy the same PDEs (19). If the same solution in (22) is chosen, the new outer loop with the state l^a and v^a is

$$\dot{l}^a = L_{vl}^a v^a + L_{\eta 2}^a R_c^T E_3 \dot{\psi} \tag{27a}$$

$$\dot{v}^a = \left[\dot{\psi} R_c E_3\right]_\times v^a + R_c R_{\theta\phi} g_1\left(F^c, \eta\right) \tag{27b}$$

where $L_{vl}^a = \begin{bmatrix} \mu_\alpha^a \lambda s_{\alpha^a} & \mu_\alpha^a \lambda c_{\alpha^a} & -\mu_\alpha^a \rho^a \\ \mu_\rho^a \lambda s_\alpha^a & \mu_\rho^a \lambda c_{\alpha^a} & -\mu_{\rho^a} \rho^a \end{bmatrix}$, $L_{\eta_2}^a = \begin{bmatrix} \rho^a s_{\alpha^a}/\lambda & \rho^a c_{\alpha^a}/\lambda & 1 \\ (\lambda + (\rho^a)^2/\lambda)\, c_{\alpha^a} & -(\lambda + (\rho^a)^2/\lambda)\, s_{\alpha^a} & 0 \end{bmatrix}$, $\mu_\alpha^a = -(A_i^a c_{\alpha^a} - B_i^a s_{\alpha^a})/(D_i^a \lambda)$, $\mu_\rho^a = (A_i^a \rho s_\alpha + B_i^a \rho c_{\alpha^a} + C_i^a \lambda)/(D_i^a \lambda)$, $[A_i^a, B_i^a, C_i^a]^T = R_c R_{\theta\phi}[A_i, B_i, C_i]^T$, $D_i^a = D_i$. It can be shown that if R_c is chosen as a function of ψ that makes $A_i^a = B_i^a = 0$, we can simplify (27a) by setting $\mu_\alpha^a = 0$, $\mu_\rho^a = -1/X_3^a$, where X_3^a is the depth of any point in the line.

3.4. Moment features for visual servoing

If a set of 3D target points or parallel lines is located in a plane, general and useful features for visual servoing are the moment features given in Chaumette (2004). If the orientation of the target plane in $\mathcal{N}$ is known, R_c can be taken as the product of R_{n_3} and the rotation matrix representing the orientation of the plane in $\mathcal{N}$. The motion of the new states for both point image coordinates p^a and image line parametrisation l^a is independent of the yaw motion. Point image moments in Chaumette (2004) and line image moments in Xie et al. (2014) for parallel lines can be used for regulating the translational motion of the vehicle. However, the feature to regulate the yaw motion of the vehicle for a non-horizontal plane remains

an open problem. One particular case is for horizontal target planes where R_c can be set to the identity matrix in (21), (22), (26) and the feature for regulating the yaw motion can be found according to Chaumette (2004).

Assumption 1: *The target is planar and parallel to the $n_1 - n_2$ plane. The target consists of either $N > 1$ points or lines.*

As in Jabbari et al. (2014), the moment feature for points adopted here is

$$s_{lp} = [p_{1g}, p_{2g}]^T s_{hp} \tag{28a}$$

$$s_{hp} = \sqrt{a^*/a} \tag{28b}$$

$$s_{\psi p} = \arctan\left(2\mu_{11}/(\mu_{20} - \mu_{02})\right)/2 \tag{28c}$$

where $p_{1g} = \frac{1}{N}\sum_{k=1}^{N} p_{1k}^a$, p_{ij}^a is the ith component of the jth point, $p_{2g} = \frac{1}{N}\sum_{k=1}^{N} p_{2k}^a$, $a = \mu_{20} + \mu_{02}$, $\mu_{ij} = \sum_{k=1}^{N}(p_{1k}^a - p_{1g}^a)^i(p_{2k}^a - p_{2g}^a)^j$, and a^* is the value of a when the vehicle is in its desired pose. Here, the feature s_{lp} is used to control the lateral motion of the vehicle, s_{hp} is used for height control, and $s_{\psi p}$ is for controlling the yaw motion. The kinematics using features (28) is

$$\dot{s}_{lp} = -\frac{\lambda}{X_3^{a*}}\begin{bmatrix} v_1^a \\ v_2^a \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} s_{lp}\dot{\psi} \tag{29a}$$

$$\dot{s}_{hp} = -v_3^a/X_3^{a*} \tag{29b}$$

$$\dot{s}_{\psi p} = -\dot{\psi} \tag{29c}$$

As in Xie et al. (2014), the selected moment features for a set of parallel lines is

$$s_{ll} = \rho_g s_{hl} \tag{30a}$$

$$s_{hl} = \sqrt{\mu_2^*/\mu_2} \tag{30b}$$

$$s_{\psi l} = \sum_{k=1}^{N} \alpha_k^a \tag{30c}$$

where $\rho_g = \sum_{k=1}^{N} \rho_k^a$, ρ_k^a is the offset of the kth line, $\mu_2 = \sum_{k=1}^{N}\left(\rho_k^a - \rho_g\right)^2$, μ_2^* is the desired value of μ_2 when the vehicle is at its desired pose, and α_k is the angle of kth line. The kinematics using features (30) is

$$\dot{s}_{ll} = -\frac{\lambda}{X_3^{a*}}\left[s_\alpha^a \;\; c_\alpha^a\right]\begin{bmatrix} v_1^a \\ v_2^a \end{bmatrix} \tag{31a}$$

$$\dot{s}_{hl} = -v_3^a/X_3^{a*} \tag{31b}$$

$$\dot{s}_{\psi l} = \dot{\psi} \tag{31c}$$

4. Controller design

The objective of IBVS is to minimise an image feature error e_s. In this section, we propose a PID controller for the outer-loop based on the image moment features given in Section 3. The control design considers system uncertainties and a stability proof is given.

4.1. Uncertainty modelling

During the identification experiments of the UAV's thrust model, we observed a slow decrease in thrust gain K_T as the battery voltage dropped during flight. In practice, the effect of decreased K_T is significant as the vehicle looses altitude. To compensate for this variation, we treat K_T as an unknown constant parameter. We remark that accounting for variation in K_T is equivalent to dealing with uncertainty in mass m as the parameters enter as a ratio in the dynamics (1a). We also observed from our previous results in Fink et al. (2015) that the image feature error converges to a non-zero constant. After an exhaustive search for the source of this error, we concluded the problem is due to Euler angle estimate bias in the AHRS output. Even small bias (i.e., 1°) leads to significant steady-state error. To account for this error, we model $R_{\theta\phi}$ as $R_{\theta\phi} = R_{n_1}^T(\phi_e)R_{n_2}^T(\theta_e)R_{n_2}(\theta_m)R_{n_1}(\phi_m)$, where the bias errors ϕ_e, θ_e are taken as constants, and the ϕ_m, θ_m are measured roll and pitch angles.

We observe from our previous work Fink et al. (2015), the reference roll-pitch angles have to be limited to a small range to avoid the target leaving the camera's FoV. Given the small range of ϕ and θ, we make a small angle approximation in the force model F^a given by (23b) as

$$\begin{aligned} F^a &= R_{n_1}^T(\phi_e)R_{n_2}^T(\theta_e)R_{n_2}(\theta_m)R_{n_1}(\phi_m)\left(gR^TE_3 + F^c/m\right) \\ &\approx \frac{K_T u}{m}\begin{bmatrix} -\theta_m + \theta_e \\ \phi_m - \phi_e \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \end{aligned} \tag{32}$$

It can be seen from (32) that the bias errors ϕ_e, θ_e enter as additive input disturbances to the outer loop, which causes the image feature errors to converge to non-zero constants.

The bias angles also create errors using the state transformation approach. We derive the relation between bias angles and image feature transformation error. We consider a 3D point in $\mathcal{C}$, $P^c = [X_1, X_2, X_3]^T$. We can express P^c in the ideal virtual camera coordinates as

$$P^{a*} = R_{n_1}^T(\phi_e)R_{n_2}^T(\theta_e)R_{n_2}(\theta_m)R_{n_1}(\phi_m)P^c \tag{33}$$

and in the real virtual camera plane coordinates as

$$P^a = R_{n_2}(\theta_m)R_{n_1}(\phi_m)P^c \tag{34}$$

Hence, from (33) and (34) we have

$$P^a = R_{n_2}(\theta_e)R_{n_1}(\phi_e)P^{a*}$$

Assuming small bias angles, we have

$$R_{n_2}(\theta_e)R_{n_1}(\phi_e) = \begin{bmatrix} c_{\theta_e} & s_{\theta_e}s_{\phi_e} & s_{\theta_e}c_{\phi_e} \\ 0 & c_{\phi_e} & -s_{\phi_e} \\ -s_{\theta_e} & c_{\theta_e}s_{\phi_e} & c_{\theta_e}c_{\phi_e} \end{bmatrix} \approx \begin{bmatrix} 1 & 0 & \theta_e \\ 0 & 1 & -\phi_e \\ -\theta_e & \phi_e & 1 \end{bmatrix}$$

Letting $P^{a*} = [X_1^{a*}, X_2^{a*}, X_3^{a*}]^T$, we obtain the dependence of P^a on the components of P^{a*} and bias angles as

$$P^a \approx \begin{bmatrix} 1 & 0 & \theta_e \\ 0 & 1 & -\phi_e \\ -\theta_e & \phi_e & 1 \end{bmatrix} \begin{bmatrix} X_1^{a*} \\ X_2^{a*} \\ X_3^{a*} \end{bmatrix} = \begin{bmatrix} X_1^{a*} + \theta_e X_3^{a*} \\ X_2^{a*} - \phi_e X_3^{a*} \\ -\theta_e X_1^{a*} + \phi_e X_2^{a*} + X_3^{a*} \end{bmatrix}$$

Since θ_e and ϕ_e are small and in practise we can assume $X_1^{a*} \ll X_3^{a*}, X_2^{a*} \ll X_3^{a*}$, and the image coordinates of the point P in real camera are approximated as

$$\begin{bmatrix} p_1^a \\ p_2^a \end{bmatrix} \approx \lambda \begin{bmatrix} \dfrac{X_1^{a*} + \theta_e X_3^{a*}}{-\theta_e X_1^{a*} + \phi_e X_2^{a*} + X_3^{a*}} \\ \dfrac{X_2^{a*} - \phi_e X_3^{a*}}{-\theta_e X_1^{a*} + \phi_e X_2^{a*} + X_3^{a*}} \end{bmatrix} \approx \lambda \begin{bmatrix} \dfrac{X_1^{a*} + \theta_e X_3^{a*}}{X_3^{a*}} \\ \dfrac{X_2^{a*} - \phi_e X_3^{a*}}{X_3^{a*}} \end{bmatrix} = \begin{bmatrix} p_1^{a*} + \lambda\theta_e \\ p_2^{a*} - \lambda\phi_e \end{bmatrix}$$

Hence, $p_{1g} = p_{1g}^* + \lambda\theta_e$, $p_{2g} = p_{2g}^* - \lambda\phi_e$, $s_{lp} = s_{lp}^* + \lambda\begin{bmatrix} \theta_e \\ -\phi_e \end{bmatrix}$, $s_{hp} = s_{hp}^*$, and $s_{\phi p} = s_{\phi p}^*$. This implies there is a small and almost constant error between the features in the real virtual camera and the ideal virtual camera under the following assumption

Assumption 2: *The bias angles θ_e, ϕ_e are small and the altitude of the vehicle is large relative to its lateral displacement.*

The kinematics of the real virtual camera is (23) with $R_c = R_{n_2}(\theta_e)R_{n_1}(\phi_e)$ and the kinematics of an ideal virtual camera is (23) with $R_c = I$. Under Assumption 2, we have $R_{n_2}(\theta_e)R_{n_1}(\phi_e) \approx I$ and hence the kinematics of the real and ideal virtual camera are almost the same.

4.2. Yaw motion controller

For both the point and line case, the kinematics of the feature for yaw motion of the vehicle is (29c) and (31c). The goal is to regulate $s_{\psi p}$ and $s_{\psi l}$ to zero. Thus, the image errors are $e_{s_{\psi p}} = s_{\psi p}$ and $e_{s_{\psi l}} = s_{\psi l}$. Evidently, the controller

$$\dot{\psi} = K_\psi e_{s_{\psi p}} \tag{35}$$

for point features and

$$\dot{\psi} = -K_\psi e_{s_{\psi l}}$$

for line features, leads to closed-loop dynamics

$$\dot{e}_{s_{\psi p}} = -K_\psi e_{s_{\psi p}} \tag{36}$$

or

$$\dot{e}_{e_{\psi l}} = -K_\psi e_{s_{\psi l}} \tag{37}$$

We can easily conclude that the closed-loop dynamics (36) or (37) is exponentially stable.

4.3. Height controller

The desired features s_{hp} and s_{hl} are set to 1. The image errors for both feature are defined as $e_{s_{hp}} = s_{hp} - 1$ and $e_{s_{hl}} = s_{hl} - 1$. The height subsystems using points and lines are the same. Thus, below we use s_h to denote both s_{hp} and s_{hl}. The image error of s_h is denoted $e_{s_h} = s_h - 1$. According to (29b), (31b), (23b), and (32), the open-loop height subsystem is

$$\dot{e}_{s_h} = -v_3^a / X_3^{a*} \tag{38a}$$

$$\dot{v}_3^a = g - K_T u/m \tag{38b}$$

We propose a PID controller

$$u = -K_{hp} e_{s_h} + K_{hd} v_3 - K_{hi}\xi_h \tag{39}$$

where $\xi_h = \int_0^t e_{s_h}(\varepsilon)d\varepsilon$. Defining $\bar{\xi}_h = g/K_{hi} + \xi_h$, the closed-loop from (38) and (39) is

$$\begin{bmatrix} \dot{\bar{\xi}}_h \\ \dot{e}_{s_h} \\ \dot{v}_3^a \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1/X_3^{a*} \\ K_T K_{hi}/m & K_T K_{hp}/m & -K_T K_{hd}/m \end{bmatrix} \begin{bmatrix} \bar{\xi}_h \\ e_{s_h} \\ v_3^a \end{bmatrix} \tag{40}$$

It can be shown that when $K_{hp}, K_{hd}, K_{hi} > 0$ and satisfy

$$K_{hi} < K_{hp} K_{hd} K_T / m \tag{41}$$

the origin of closed-loop system (40) is exponentially stable.

4.4. Lateral controller design

Without loss of generality, the desired feature values for s_{lp} and s_{ll} are set to zero. Hence, the image errors for lateral motion are $e_{s_{lp}} = [e_{s_{lp1}}, e_{s_{lp2}}]^T = s_{lp}$ and $e_{s_{ll}} = [e_{s_{ll1}}, e_{s_{ll2}}]^T = s_{ll}$.

4.4.1. Point moment feature

According to (29a), (23b), and (32), the open-loop lateral subsystem in the transformed state coordinates is

$$\dot{e}_{s_{lp}} = -\frac{\lambda}{X_3^{a*}} v_l^a - \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} e_{s_{lp}} \dot{\psi} \tag{42a}$$

$$\dot{v}_l^a = \frac{K_T u}{m} \begin{bmatrix} -\theta_m + \theta_e \\ \phi_m - \phi_e \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} v_l^a \dot{\psi} \tag{42b}$$

where $v_l^a = [v_1^a, v_2^a]^T$. Similarly, we propose a PID controller as

$$u \begin{bmatrix} \theta_m \\ -\phi_m \end{bmatrix} = -K_{lp} e_{s_{lp}} + K_{ld} v_l^a - u K_{li} \xi_{lp} \tag{43}$$

where $\xi_l = \int_0^t e_{s_{lp}}(\varepsilon) d\varepsilon$. Defining u_g as the value of thrust which compensates gravity, and $\bar{\xi}_{lp} = \xi_{lp} + \frac{1}{K_{li}} \begin{bmatrix} \theta_e \\ -\phi_e \end{bmatrix}$, we obtain the closed-loop system

$$\dot{x}_{lp} = A_{cl} x_{lp} + D(t) x_{lp} \tag{44}$$

where $x_{lp} = [\bar{\xi}_{lp}^T, e_{s_{lp}}^T, (v_l^a)^T]^T$, $A_{cl} = \begin{bmatrix} 0 & I & 0 \\ 0 & 0 & -\lambda I/X_3^{a*} \\ K_T u_g K_{li} I/m & K_T K_{lp} I/m & -K_T K_{ld} I/m \end{bmatrix}$, $D(t) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\dot{\psi} S & 0 \\ K_T \tilde{u} K_{li} I/m & 0 & -\dot{\psi} S \end{bmatrix}$, $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\tilde{u} = u - u_g$. If the PID control parameters satisfy

$$K_{li} < K_T K_{lp} K_{ld} / (u_g m) \tag{45}$$

then A_{cl} is Hurwitz. Because the closed-loops for height and yaw are exponentially stable, we have $\dot{\psi} \to 0$ and $\tilde{u} \to 0$. This implies $D(t)x \to 0$. Using a converse Lyapunov theorem in Khalil (2001, Thm. 4.14), we can conclude that the closed-loop (44) is exponentially stable.

4.4.2. Line moment feature

Based on (31a), (23b), and (32), the lateral subsystem is

$$\dot{e}_{s_{ll}} = -\lambda v_2^a / X_3^{a*} - \lambda \left(s_\alpha^a v_1^a + \left(c_\alpha^a - 1 \right) v_2^a \right) / X_3^{a*} \tag{46a}$$

$$\dot{v}_2^a = K_T u \left(\phi_m - \phi_e \right) / m - v_1^a \dot{\psi} \tag{46b}$$

A PID controller is selected as

$$u \phi_m = K_{lp} e_{s_{ll}} - K_{ld} v_2^a + u K_{li} \xi_{ll} \tag{47}$$

where $\xi_{ll} = \int_0^t e_{s_{ll}}(\varepsilon) d\varepsilon$. Define $\bar{\xi}_{ll} = -\phi_e / K_{li} + \xi_{ll}$, the closed-loop becomes

$$\dot{x}_l = A_{cl} x_l + D_l(t) \tag{48}$$

where $x_{ll} = [\bar{\xi}_{ll}, e_{s_{ll}}, v_2^a]^T$, $A_{cl} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -\lambda/X_3^{a*} \\ K_T u_g K_{li}/m & K_T K_{lp}/m & -K_T K_{ld}/m \end{bmatrix}$, $\tilde{u} = u - u_g$, $D_l(t) = \begin{bmatrix} 0 \\ \lambda \left(s_\alpha^a v_1^a + \left(c_\alpha^a - 1 \right) v_2^a \right) / X_3^{a*} \\ K_T \tilde{u} K_{li} \bar{\xi}_{ll} / m - v_1^a \dot{\psi} \end{bmatrix}$. Since the closed loop for height and yaw are exponentially stable, we have $\alpha \to 0$, $\tilde{u} \to 0$, $\dot{\psi} \to 0$, $D_l(t) \to 0$. Using the results in Marino and Tomei (1995, Lemma III.1), we can conclude that if the PID control gains satisfy (45), then (48) is exponentially stable.

4.4.3. Inner–outer loop stability analysis

If the inner loop tracks its reference provided from the outer loop (43) or (47), then stability has been proven above. In this section, we consider the effect of tracking error from the inner loop on the entire closed-loop stability.

Denote $e_{\eta_1} = \eta_{m_1} - \eta_{m_1}^*$, $\eta_{m_1} = [\phi_m, \theta_m]^T$, $\eta_{m_1}^* = [\phi_m^*, \theta_m^*]^T$, and $\eta_{e_1} = [\phi_e, \theta_e]^T$, the open-loop lateral subsystem (42) for point features becomes

$$\dot{e}_{s_{lp}} = -\frac{\lambda}{X_3^{a*}} v_l^a - S e_{s_{lp}} \dot{\psi} \tag{49a}$$

$$\dot{v}_l^a = \frac{K_T u}{m} S \left(\eta_{m_1}^* - \eta_{e_1} \right) - S v_l^a \dot{\psi} + \frac{K_T u}{m} S e_{\eta_1} \tag{49b}$$

The desired attitude is chosen as

$$u S \eta_{m_1}^* = K_{lp} e_{s_{lp}} - K_{ld} v_l^a + u K_{li} \xi_{lp} \tag{50}$$

The outer loop becomes

$$\dot{x}_{lp} = A_{cl} x_{lp} + B_{lp} e_{\eta_1} + D_{lp}(t) \tag{51}$$

where $B_{lp} = [0_2, 0_2, \frac{K_T u_g}{m} S^T]^T$, $D_{lp} = D(t) x_{lp} + B_u(t) e_{\eta_1}$, $B_u = [0_2, 0_2, \frac{K_T \tilde{u}}{m} S^T]^T$. Since the yaw ψ and u exponentially converge to constants, the term D_{lp} exponentially converges to zero. From (43) we have

$$\dot{u} S \eta_{m_1}^* + u S \dot{\eta}_{m_1}^* = K_l \dot{x}_{lp} + \dot{u} K_{li} \xi_{lp} \tag{52}$$

where $K_l = [uK_{li}I_2,\ K_{lp}I_2,\ -K_{ld}I_2]$. Substituting (44) into (52), we obtain

$$\dot{\eta}^*_{m_1} = \frac{1}{u_g}S^{-1}K_lA_{cl}x_{lp} + \frac{1}{u_g}S^{-1}K_lB_{lp}e_{\eta_1} + D^*_{\eta_{m_1}}(t) \quad (53)$$

where

$$D^*_{\eta_{m_1}}(t) = \frac{1}{u}S^{-1}(K_lD(t)x_{lp} + K_lB_u(t)e_{\eta_1} + \dot{u}\left(K_{li}\xi_{lp} - S\eta^*_{m_1}\right)) + \frac{\tilde{u}}{uu_g}S^{-1}(K_lA_{cl}x_{lp} + B_{lp}e_{\eta_1})$$

The vector $D^*_{\eta_{m_1}}$ exponentially converges to zero since $\dot{u}$, $\tilde{u}$, D, B_u exponentially converge to zero.

Motivated by Castillo, Lozano, and Dzul (2005), we approximate the inner loop as

$$\dot{e}_{\eta_1} = \dot{\eta}_{m_1} - \dot{\eta}^*_{m_1} \quad (54)$$

$$\ddot{\eta}_{m_1} = J_{12}^{-1}\tau^c_{12} \quad (55)$$

where we have assumed a diagonal inertia matrix with nonzero components J_i, $i = 1, 2, 3$, $J_{12} = \mathrm{diag}([J_1, J_2])$, and τ^c_{12} is a vector of the first two components of τ^c.

Following Kendoul, Fantoni, and Lozano (2008), one control design is

$$\tau^c_{12} = \ddot{\eta}^*_{m_1} - K_{\eta_1 p}e_{\eta_1} - K_{\eta_1 d}\dot{e}_{\eta_1} \quad (56)$$

In this case, we can show that the control gains for the inner and outer loops can be independently chosen to ensure stability. However, computing the first and second derivative of the reference angle can introduce noise and is often omitted in practice (Bouabdallah, Murrieri, & Siegwart, 2004). Noise is especially of concern in visual servoing where the reference angle depends on noisy image features.

Defining $\delta_1 = \int_0^t e_{\eta_1}dt$, $\delta_2 = \frac{e_{\eta_1}}{k_1} + \delta_1$, $\delta_3 = \frac{\dot{\eta}_{m_1}}{k_2} + \delta_2$, $x_\delta = [\delta_1^T, \delta_2^T, \delta_3^T]^T$, and choosing the control law as $\tau^c_{12} = -k_3\delta_3$, we have

$$\dot{x}_\delta = A_\delta x_\delta + \frac{1}{k_1}B_\delta x_{lp} + D_\delta(t) \quad (57)$$

where

$$A_\eta = \begin{bmatrix} -k_1I_2 & k_1I_2 & 0 \\ -(k_1 + \frac{K_TK_{ld}}{m})I_2 & -\left(\frac{k_2-k_1^2}{k_1} - \frac{K_TK_{ld}}{m}\right)I_2 & \frac{k_2}{k_1}I_2 \\ -(k_1 + \frac{K_TK_{ld}}{m})I_2 & -\left(\frac{k_2-k_1^2}{k_1} - \frac{K_TK_{ld}}{m}\right)I_2 & -\frac{k_3}{k_2}J_{12}^{-1} + \frac{k_2}{k_1}I_2 \end{bmatrix}$$

$$B_\delta = \begin{bmatrix} 0 \\ -\frac{1}{u_g}S^{-1}K_lA_{cl} \\ -\frac{1}{u_g}S^{-1}K_lA_{cl} \end{bmatrix},\ D_\delta(t) = \frac{1}{k_1}\begin{bmatrix} 0 \\ D^*_{\eta_{m_1}}(t) \\ D^*_{\eta_{m_1}}(t) \end{bmatrix}$$

and $D_\delta(t)$ exponentially converges to zero. We define a nominal system for closed-loop (44), (57) as

$$\dot{x}_{lp} = A_{cl}x_{lp} + B_{lp}e_{\eta_1}$$
$$\dot{x}_\delta = A_\delta x_\delta + \frac{1}{k_1}B_\delta x_{lp}$$

Considering a Lyapunov function candidate as

$$V = x_{lp}^TP_lx_{lp} + \frac{1}{2}x_\delta^Tx_\delta \quad (58)$$

where $P_l > 0$, and $A_{cl}^TP_l + P_lA_{cl} = Q_l$, $Q_l > 0$, its time derivative is

$$\dot{V} = -x_{lp}^TQ_lx_{lp} + 2x_{lp}^TP_lB_{lp}e_{\eta_1} + \frac{1}{2}x_\delta^T\left(A_\eta^T + A_\eta\right)x_\delta + \frac{1}{2k_1}x_{lp}^T\left(B_\delta + B_\delta^T\right)x_\delta \quad (59)$$

Since P_l, B_{lp}, and B_δ are independent of gains k_i, $i = 1, 2, 3$, we can choose appropriate gains k_i, $i = 1, 2, 3$ that the $A_\eta^T + A_\eta < -Q_\eta$, where $Q_\eta > 0$. If the eigenvalues of Q_η are taken sufficiently large, then we can apply Young's inequality to show $\dot{V} < 0$. Since the two terms D_δ and D_{lp} are exponentially convergent, using Marino and Tomei (1995, Lemma III.1) shows the entire closed-loop is stable. We remark that the control law is given by

$$\tau^c_{12} = -k_3\delta_3 = -k_3\left(\frac{\dot{\eta}_{m_1}}{k_2} + \frac{e_{\eta_1}}{k_1} + \delta_1\right) = -K_i\delta_1 - K_pe_{\eta_1} - K_d\dot{\eta}_{m_1} \quad (60)$$

where $K_i = k_3$, $K_p = k_3/k_1$, and $K_d = k_3/k_2$. Hence, the inner loop controller can be implemented with a PID structure.

5. Experimental work

In this section, we briefly introduce the experimental platform used to validate the performance of the state transformation based dynamic visual servoing. Next, we present the experimental visual servoing results for image point moment features and image line moment features.

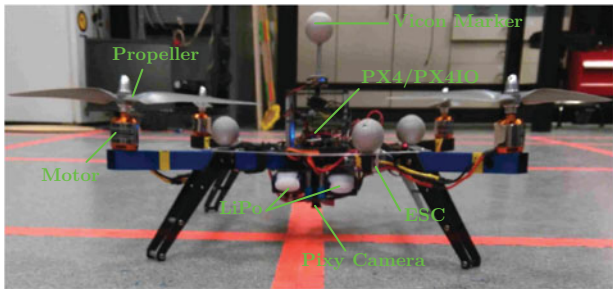


Figure 4. The ANCL quadrotor platform.

Table 1. Nominal quadrotor and camera model parameters.

m	1.6 kg
J	diag([0.03, 0.03, 0.05]) kg·m^2
λ	2.8 mm
Image sensor size	1/4″
Resolution	(320, 200) pixels
δ	0.25 m

5.1. Experimental platform

The indoor Applied Nonlinear Control Lab (ANCL) quadrotor platform shown in Figure 4 is used to experimentally validate the state transformation based dynamic visual servoing law.

The autopilot is based on a customised PX4 system which is an open-hardware and open-source project (PX4, 2014). A Kalman filter-based AHRS fuses sensor data from the on-board 3D gyroscope, 3D accelerometer, and 3D magnetometer to provide attitude and angular velocity estimates. The vision system is a low cost CMUcam5 Pixy which is mounted at the bottom of the quadrotor and sends image features to the PX4 via a serial connection. An eight camera Bonita Vicon motion capture system measures the quadrotor's pose at up to 200 Hz and sends the measurements to a ground computer. This computer calculates and sends the quadrotor's translational velocity to the PX4 via a radio. Table 1 summarises the quadrotor's and camera's nominal model parameters. See Fink et al. (2015) for more details.

Table 2. Outer loop control parameters.

Gain	K_ψ	K_{hp}	K_{hd}	K_{hi}	K_{lp}	K_{ld}	K_{li}
Value	0.6	0.09	0.1	0.045	0.08	0.22	0.02

Table 3. PID control gains for inner loop attitude tracking.

	Roll	Pitch	Yaw
K_p	0.33	0.345	0.3
K_d	0.08	0.09	0.08
K_i	0.03	0.03	0.005

5.2. Point features

The proposed controller considered in this section is (36), (39), and (43). Two visual target points are put on the horizontal ground with a displacement of 0.44 m. The control gains for the outer loop are shown in Table 2. The inner loop attitude controller is a PID controller with gains K_p, K_d, and K_i. Their values are shown in Table 3. An example of the roll and pitch tracking performance with this PID controller is shown in Figure 5. To check the performance of the integral component of the PID controller to compensate the uncertainties on K_T, m and the bias in roll and pitch angles ϕ_e and θ_e, we consider two cases: a PD controller with K_{li} in (43) and K_{hi} in (39) set to zero; a PID controller with K_{li} and K_{hi} set to values given in Table 2. The value of a^* in (28b) is set as 0.4, which is obtained from an image when the vehicle hovers at the desired height. The trajectory of $e_s = [e_{s_{lp}}^T, e_{s_{hp}}, e_{s_{\psi p}}]^T$ is shown in Figure 6, where Figure 6(a) and 6(b) gives the results with PD and PID controller, respectively. Figure 7(a) and 7(b) show the corresponding 3D translational error $e_d = [e_{d_1}, e_{d_2}, e_{d_3}]^T = d_{nc}^n - d_{nc}^{n*}$ and $e_\psi = \psi - \psi^*$ for two cases, where $d_{nc}^{n*} = [0.1080, -0.1314, -1.3927]^T$ m and $\psi^* = -0.5454$ rad correspond to the desired d_{nc}^n and ψ, respectively, when $e_{s_{lp}} = 0$, $e_{s_{hp}} = 0$, and $e_{s_{\psi p}} = 0$.

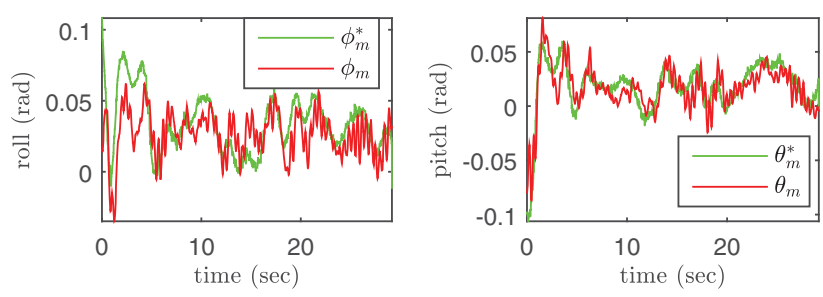


Figure 5. Roll and pitch tracking performance.

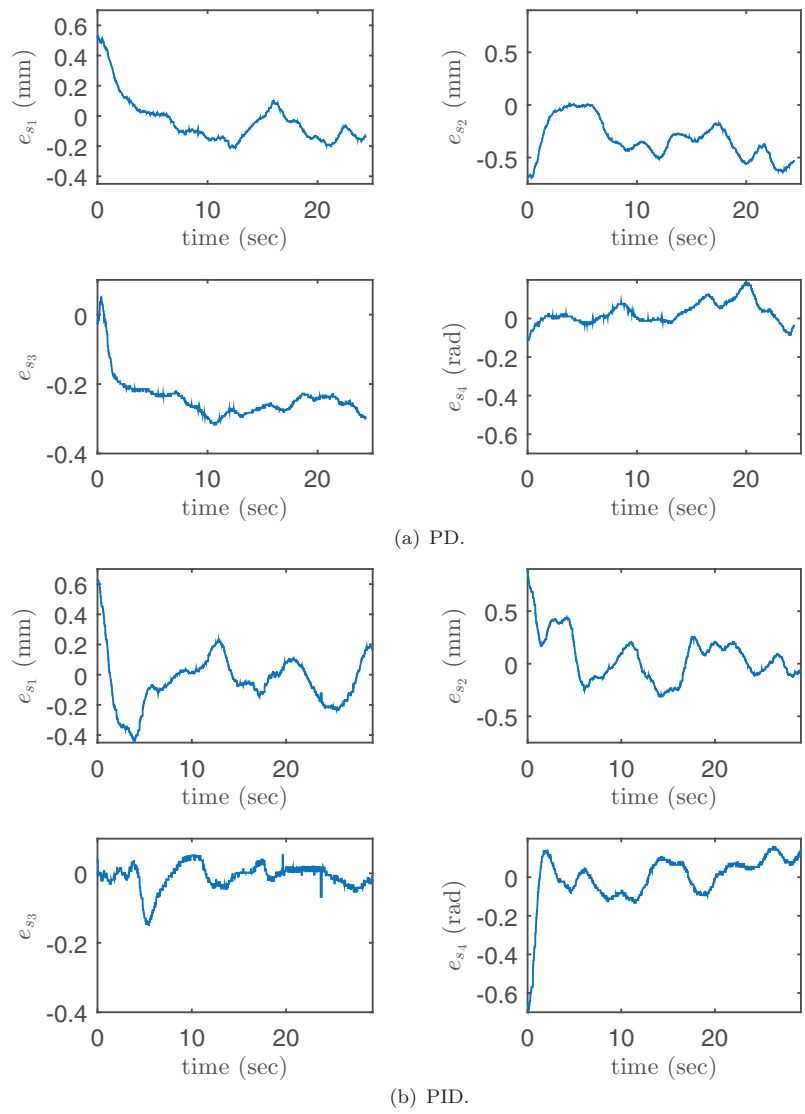


Figure 6. Experimental trajectories of image feature error e_s with point features.

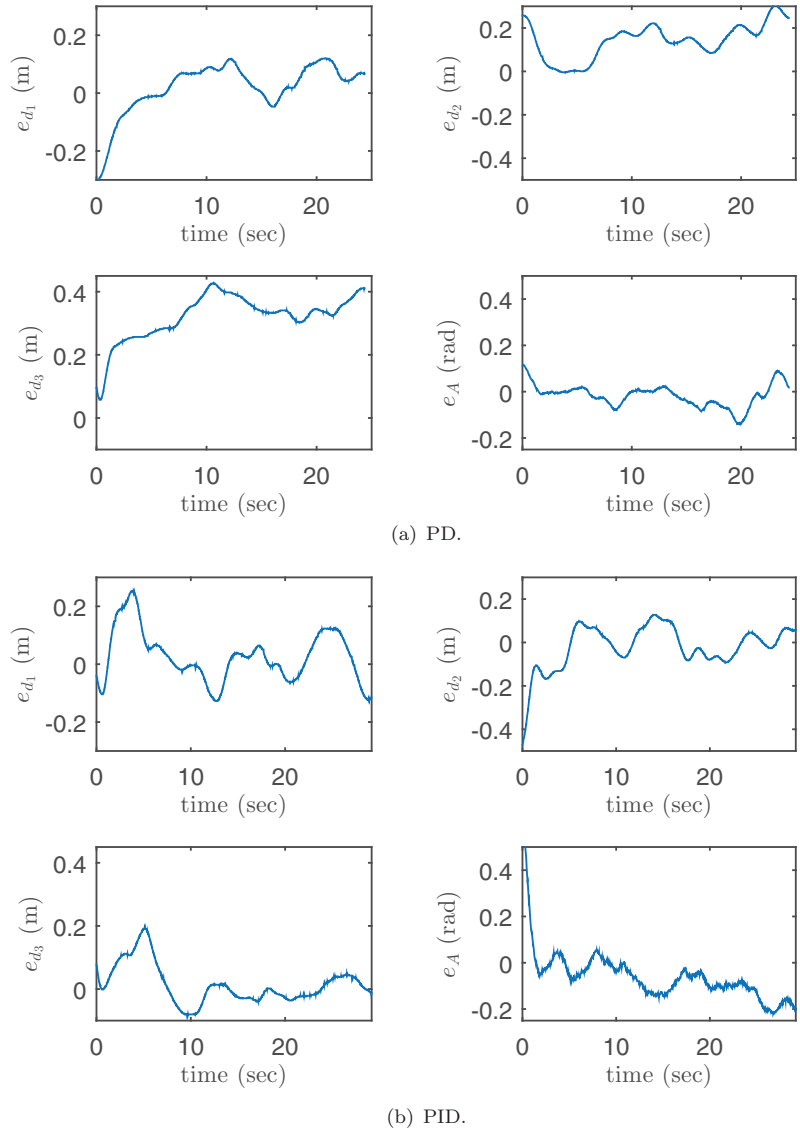
Table 4. Statistical properties of error for point feature IBVS. We denote mean as μ and standard deviation as σ.

	PID controller		PD controller	
Variable	μ	σ	μ	σ
$e_{s_{lp1}}$ (mm)	− 0.01	0.12	− 0.10	0.08
$e_{s_{lp2}}$ (mm)	0.00	0.16	− 0.40	0.13
$e_{s_{hp}}$	− 0.00	0.03	− 0.27	0.02
$e_{s_{\psi p}}$ (rad.)	0.03	0.08	0.05	0.06
e_{d_1} (m)	0.003	0.071	0.057	0.046
e_{d_2} (m)	0.005	0.065	0.179	0.056
e_{d_3} (m)	− 0.008	0.029	0.451	0.034
e_ψ (rad.)	− 0.109	0.053	− 0.026	0.053

It can be seen that both cases reach steady-state after around 10 seconds. Table 4 gives the averages and standard deviations of signals $e_{s_{lp}}$, $e_{s_{hp}}$, $e_{s_{\psi p}}$, e_d, and e_ψ after 10 seconds. It can be seen from Figure 6 and Table 4 that $e_{s_{lp}}$ and $e_{s_{hp}}$ are significantly reduced when the integral control is added. We can also tell from Figure 6 and Table 4 that e_d is reduced when the integral control is added. The performance of yaw regulation for the two cases is the same because the same control law for yaw is applied. We also observe that the standard deviations for the two cases are similar. Hence, we can conclude that the proposed PID controller can efficiently deal with the model uncertainties described in Subsection 4.1.

5.3. Line features

The proposed controller considered in this section is (37), (39), and (47). We use line moment features (31). Two parallel visual target lines with a displacement of 0.2 m are put on the ground. As with the point feature



(a) PD.

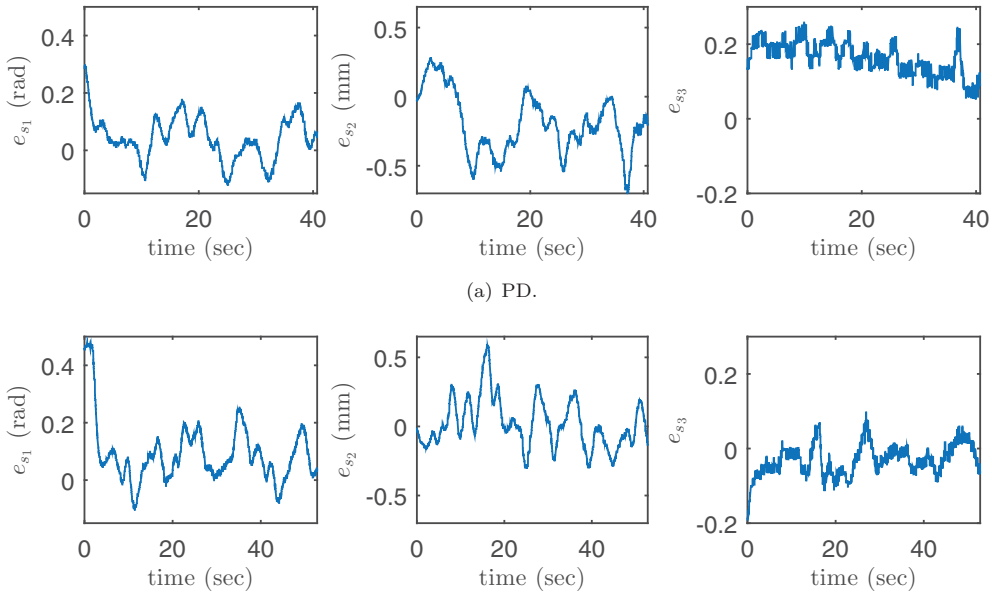
(b) PID.

Figure 7. Experimental trajectories of vehicle's 3D position error e_d and yaw error e_ψ with point features.

Table 5. Outer loop control parameters.

Gain	K_ψ	K_{hp}	K_{hd}	K_{hi}	K_{lp}	K_{ld}	K_{li}
Value	0.4	0.09	0.08	0.01	0.08	0.2	0.02

case, both PID and PD control are implemented to show the benefit of the integral term which compensates uncertainty. The control gains for the outer loop are shown in Table 5. The PID control gains for the inner loop are the same as for point features and are given in Table 3. Hence, the inner loop attitude tracking performance is similar to that shown in Figure 5. The desired value μ_2^* in (30b) is set to 0.082 which is acquired from an image when the vehicle flies at the desired height. The trajectory of image feature error $e_s = [e_{s_{ll}}, e_{s_{hl}}, e_{s_{\psi l}}]^T$ is shown in Figure 8, where Figure 8(a) shows the performance of the PD controller and Figure 8(b) shows the performance of PID controller. Since it is impossible to regulate the 3D position of the vehicle with a set of parallel lines, only partial 3D position errors $e_{d_2} = d_2 - d_2^*$, $e_{d_3} = d_3 - d_3^*$ and yaw error $e_\psi = \psi - \psi^*$ are shown in Figure 9, where $d_2^* = -0.1241$ m, $d_3^* = -1.3670$ m, $\psi^* = -0.7526$ rad are the desired lateral position in the n_2 direction, the desired height, and desired yaw, respectively. We observe from Figures 8 and 9 that the vehicle reaches its steady state after around 10 seconds. Table 6 gives the means and standard deviations of signals e_s, e_d,

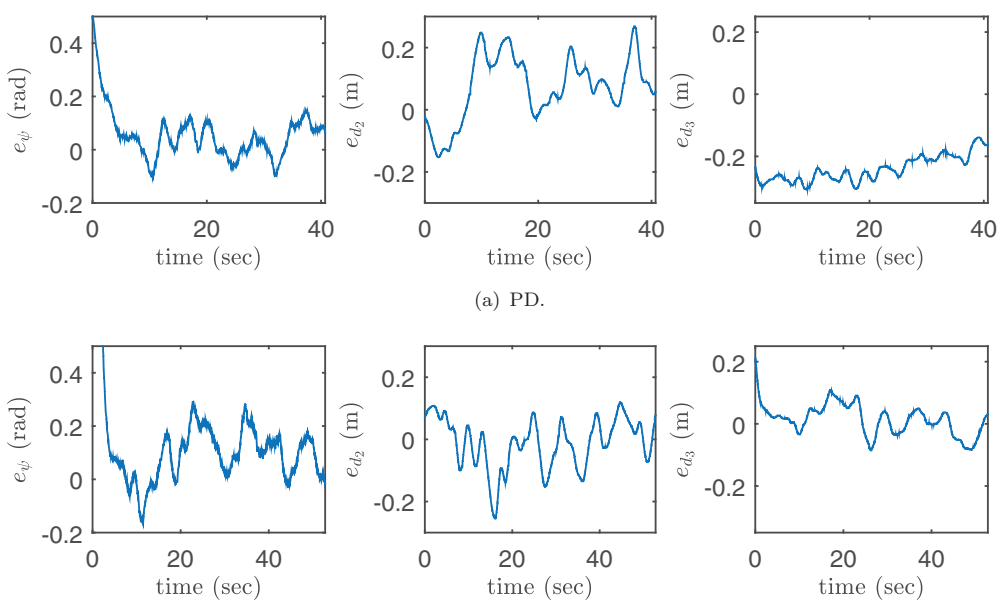


(a) PD.

(b) PID.

Figure 8. Experimental trajectories of image feature error e_s with line features.

and e_ψ. It can be seen from Figure 8 and Table 6 that when the PID controller is enabled, the means of image feature errors $e_{s_{ll}}$, $e_{s_{hl}}$ are significantly reduced relative to the PD controller case. The standard deviations of these signals for the two cases are similar. From Figure 9 and Table 6, we also observe that the means of the corresponding 3D position errors e_{d_2}, e_{d_3} are significantly reduced as well. The image error $e_{s_{\psi l}}$ and the yaw error $e_\psi = \psi - \psi^*$ for two cases is reduced to a small value from a large initial value. The mean and standard deviation of $e_{s_{\psi l}}$ and e_ψ at steady state for the two cases are similar because the same control law (37) is used.



(a) PD.

(b) PID.

Figure 9. Experimental trajectories of vehicle's partial 3D position error e_d and yaw error e_ψ with line features.

Table 6. Statistical properties of e_s, e_d, and e_ψ for line feature IBVS. We denote mean as μ and standard deviation as σ.

	PID controller		PD controller	
Variable	μ	σ	μ	σ
$e_{s_{\psi l}}$ (rad.)	0.07	0.07	0.04	0.07
$e_{s_{ll}}$ (mm)	0.03	0.18	− 0.26	0.17
$e_{s_{hl}}$	− 0.02	0.04	0.15	0.04
e_{d_2} (m)	− 0.024	0.079	0.106	0.072
e_{d_3} (m)	− 0.009	0.049	− 0.228	0.040
e_ψ (rad.)	0.093	0.091	0.041	0.056

6. Conclusion

We have presented a state transformation-based dynamic visual servoing scheme for an underactuated quadrotor UAV. The state transformation is obtained by solving a system of PDEs. A method for computing the general solutions is given. A particular solution is given by the virtual camera approach. A set of image moment features in the virtual camera is used to derive a dynamic visual servoing law. Model uncertainty including thrust gain and attitude measurement bias is considered in the visual servoing control design. The proposed PID controller is robust to this uncertainty. In addition, the PID controller does not require an estimate of depth, which is typically needed in classical IBVS. An indoor quadrotor platform is used to validate the proposed control. Experimental results demonstrate improved performance with lower mean image and position error. An analysis of roll and pitch bias on the virtual camera transformation error is provided.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix A. Procedure to solve first order linear homogeneous PDEs

A system of $q < n$ linear homogeneous PDEs in a dependent variable $f(x_1, x_2, ..., x_n)$ is defined as

$$G_i(f) = \sum_{j=0}^{n} a_{ji} \frac{\partial f}{\partial x_j} = 0, \quad 1 \le i \le q \tag{A1}$$

where G_i, $1 \le i \le q$ are vector fields, and a_{ji} is function of the n independent variables x_i, $1 \le i \le n$. According to Lee (2006), there exists $n - q$ independent solutions for (A1) if the distribution

$$\mathcal{D} = \text{span}\{G_1, G_2, ..., G_q\}$$

is q dimensional and involutive. The distribution $\mathcal{D}$ is involutive if for any two vector fields $G_i, G_j \in \mathcal{D}$, their Lie bracket $[G_i, G_j] \in \mathcal{D}$. For our application, $q = 2$ and it can be shown that the Lie brackets $[G_{1p}, G_{2p}] = 0$ where G_{ip} are defined in (17), $[G_{1v}, G_{2v}] = 0$ where G_{iv} are defined in (19), and $[G_{1l}, G_{2l}] = 0$ where G_{il} are defined in (25). Hence, the distributions $\mathcal{D}_p = \text{span}\{G_{1p}, G_{2p}\}$, $\mathcal{D}_v = \text{span}\{G_{1v}, G_{2v}\}$, $\mathcal{D}_l = \text{span}\{G_{1l}, G_{2l}\}$ are involutive. We also say the PDEs are complete for the special case where the Lie brackets are zero. Hence, there exist two independent solutions to PDEs (17), (25), and three independent solutions to (19).

Next, we outline the method to solve (A1) which is assumed complete, i.e., G_i, $1 \le i \le q$ are independent and the Lie bracket $[G_i, G_j] = 0$, $1 \le i, j \le q$. If $y_2, ..., y_n$ are independent integrals of the $G_k(f) = 0$, and choose a function y_1 such that $y_1, y_2, \ldots, y_n$ define new independent variables and $G_k(f) = 0$, for some k, reduces to $\partial f/\partial y_1 = 0$. The PDEs expressed in the y-coordinates is

$$L_i(f) = \begin{cases} \dfrac{\partial f}{\partial y_1} = 0, & i = k \\ \displaystyle\sum_{j=2}^{n} b_{ji} \frac{\partial f}{\partial y_j} = 0, & i \neq k \end{cases} \tag{A2}$$

where b_{ji} is a function of the $n - 1$ independent variables $y_2, y_3, ..., y_n$. Hence, $L_i(f) = 0$, $i \neq k$ forms a complete system of $q - 1$ equations and $n - 1$ independent variables. By repeating this procedure, the system can be reduced to one equation in $n - q + 1$ independent variables. The general solution of this equation is an arbitrary smooth function of $n - q$ independent solutions. We illustrate this procedure by solving (17). It can be easily obtained that the general integral of $G^a_{1p}(p^a_i) = 0$ is an arbitrary smooth function of $(p_2 s_\phi + \lambda c_\phi)/p_1$, $(p_2 c_\phi - \lambda s_\phi)/p_1$, and θ. We define a new set of independent variables $p_{11} = (p_2 s_\phi + \lambda c_\phi)/p_1$, $p_{21} = (p_2 c_\phi - \lambda s_\phi)/p_1$, ϕ, θ. The PDEs (17) become

$$\frac{\partial p^a_{i1}}{\partial \phi} = 0 \tag{A3a}$$

$$\frac{\partial p^a_{i1}}{\partial p_{11}} \left(p^2_{11} + 1\right) + \frac{\partial p^a_{i1}}{\partial p_{21}} p_{11} p_{21} + \frac{\partial p^a_{i1}}{\partial \theta} = 0 \tag{A3b}$$

where the new dependent variable is denoted $p_{i1}^a(p_{11}, p_{21}, \phi, \theta)$. Hence, we can conclude that p_{i1}^a is not a function of ϕ. The PDE (A3b) has three independent variables p_{11}, p_{21}, and θ. The general integral of (A3b) is an arbitrary smooth function of $(s_\theta - p_{11}c_\theta)/p_{21}$ and $(c_\theta + p_{11}s_\theta)/p_{21}$. Hence, expressed in the original variables p_1, p_2, ϕ, and θ, the general integral of (17) is

$$\kappa_2\left(\frac{s_\phi c_\theta p_2 + \lambda c_\phi c_\theta - p_1 s_\theta}{\lambda s_\phi - p_2 c_\phi}, -\frac{s_\phi s_\theta p_2 + c_\phi s_\theta \lambda + p_1 c_\theta}{\lambda s_\phi - p_2 c_\phi}\right)$$